

Automated Problem-to-Solution Generation for a Tokamak Fusion-Plasma Simulation

A hierarchical multi-agent PETSc system, applied and verified

Argonne National Laboratory - summer 2026 intern event

Why: fusion energy needs trustworthy simulation

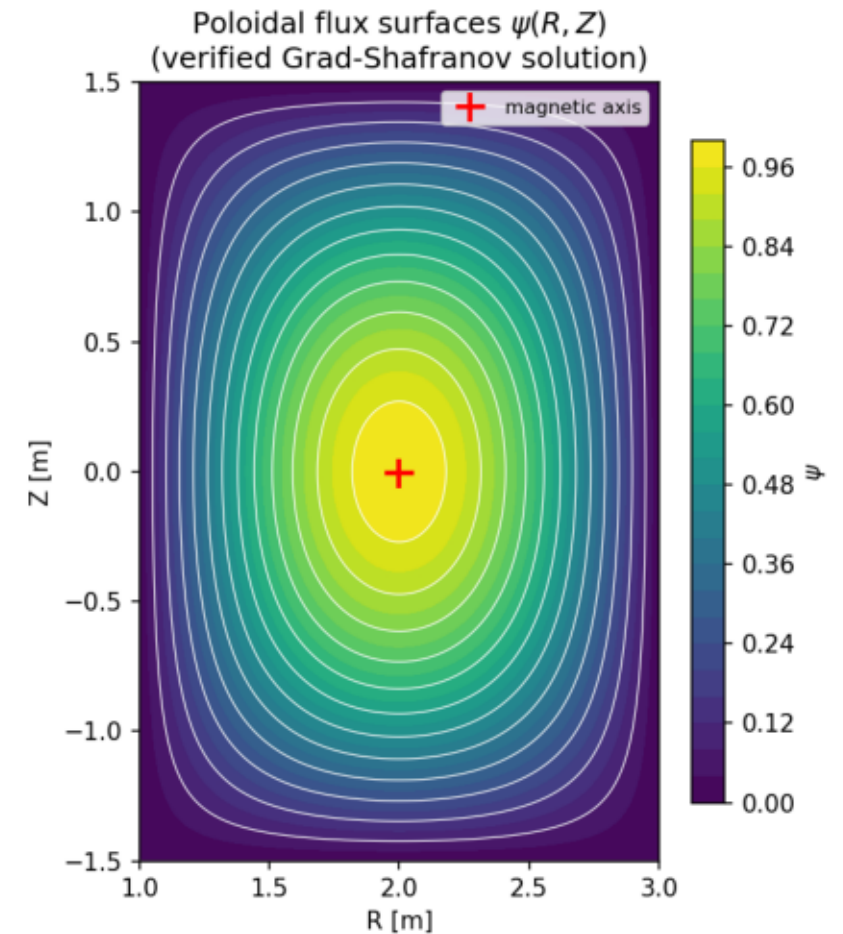
- A tokamak confines a ~ 100 -million-C plasma in a magnetic “cage” so nuclei fuse.
- Simulation predicts - before building hardware - whether the plasma stays confined.
- But turning a scientific idea into correct, fast HPC code needs scarce, implicit expertise:
 - numerical methods (grids, discretizations, solvers)
 - library APIs and parallelism (here: PETSc on many cores/GPUs)
 - verification: is the answer actually right?
- Question: can a multi-agent AI system automate this path - and can we trust the result?

The problem: tokamak Grad-Shafranov equilibrium

- Axisymmetric ideal-MHD force balance for the poloidal flux $\psi(R, Z)$:

$$-\Delta^* \psi = -\mu_0 R^2 p'(\psi) - F F'(\psi)$$

- Its solution gives the flux surfaces, magnetic axis, and safety factor q .
- Elliptic, nonlinear, time-independent $\rightarrow$ a natural PETSc SNES problem.
- Verifiable: a manufactured exact solution gives a rigorous convergence test.



The system: a 3-layer multi-agent pipeline (PETSc MCP)

- Problem Definition - Mathematical Modeling agent → strong/weak form, name, time-dep.
- Agent Execution - Numerical Analysis agent → grid, discretization, solver (SNES).
- Agent Execution - HPC Code Generation agent → writes, compiles & runs PETSc C.
- Execution substrate - compile-run agent (make / run on the real machine).
- All agents run against Argonne's Argo gateway (Claude Opus 4.8).
- A project-owned driver records every artifact with provenance (reproducible, resumable).

The pipeline in action (this run)

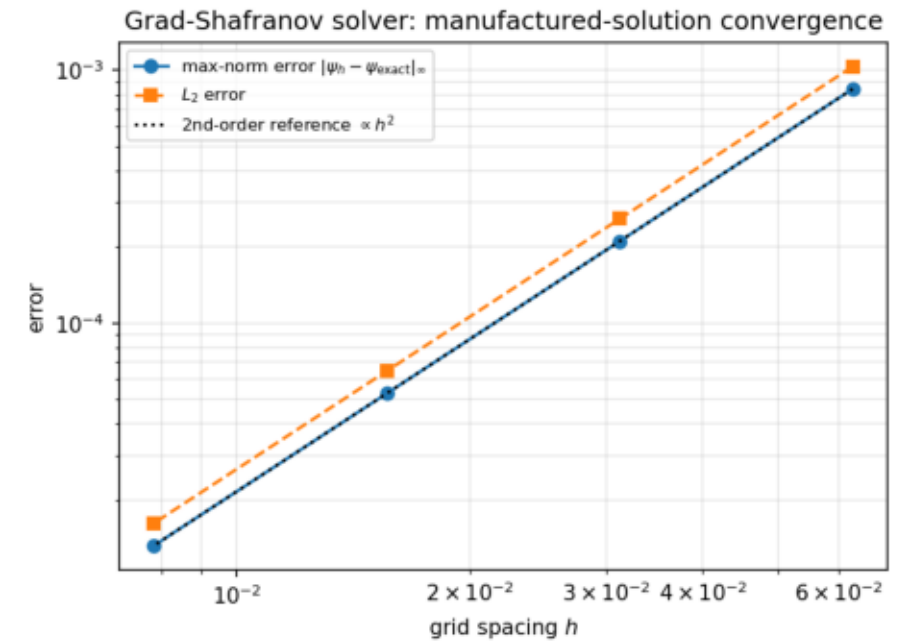
- Prompt: “the Grad-Shafranov equilibrium for the plasma in a tokamak.”
- Modeling agent → identified 'Grad-Shafranov equation'; time-dependent = False.
- Numerical Analysis agent → nonlinear solve with SNES on a structured grid.
- Code Generation agent → 267-line PETSc program (DMDA + SNES + true Jacobian).
- Compiled and ran on 1 and 4 MPI ranks with no human edits.

The agent-generated PETSc solver (excerpt)

```
static PetscErrorCode FormFunction(SNES snes, Vec X, Vec F, void *ctx){
    /* Delta*_h psi = (psi_E-2psi_C+psi_W)/hR^2
        - (1/R)(psi_E-psi_W)/(2hR)
        + (psi_N-2psi_C+psi_S)/hZ^2 */
    f[j][i] = dRR - dR1/R + dZZ - ForcingF(user,R,Z); /* residual */
}
SNESSetFunction(snes, r, FormFunction, &user);
SNESSetJacobian(snes, J, J, FormJacobian, &user); /* true Jacobian */
SNESolve(snes, NULL, x);
```

Verification: it is actually right

- Method of manufactured solutions: known exact ψ , check the numerical error.
- Max-norm error = $1.32\text{e-}05$ on 65×65 .
- Observed order of accuracy $p = 2.00, 2.00, 2.00$ (expected 2 for central differences).
- SNES: CONVERGED_FNORM_RELATIVE.



Manufactured-solution convergence

Decision-gate metrics

- Correctness: model identified ✓, compiled & ran ✓, 2nd-order convergence ✓.
- Human effort: 0 lines of solver code hand-written (it was generated).
- Efficiency: total wall-clock ≈ 450.8 s; code-gen used 5 tool calls.
- Approx. 23 LLM completions across the whole pipeline.

What I built and contributed

- A project-owned orchestration driver with full artifact provenance (resumable).
- Verification + post-processing (convergence, flux surfaces) and a metrics harness.
- Fixes contributed back to the open multi-agent system:
 - CWD-independent server resolution (absolute paths)
 - graceful operation without the documentation/RAG services
 - a code-generation loop that reliably captures results

Takeaways & next steps

- A multi-agent system generated a correct, verified PETSc fusion-MHD solver from a prompt.
- Verification-driven: convergence + exact-solution checks, not just plausible output.
- Next: shaped real-machine equilibria (Solov'ev/Cerfon-Freidberg), q-profile, GPU scale-up.
- Next: demonstrate the fully-autonomous built-in orchestrator agent end to end.
- Code + docs: github.com/engineer-scientist/petsc_mcp_servers_tokamak